

Technical Evaluation: Hidden Camera Detection Methods

Objective: Evaluating next-step detection pipelines for the ESP32 and smartphone-browser architecture.

Performance Tradeoff Overview:

There is a near-inverse relationship between a method's detection accuracy and its implementation viability given the current hardware bill of materials (BOM). The easiest methods tend to be the least accurate, creating distinct strategic tiers for development:

- **High Accuracy / High Effort:** SweepLED (Optimal Long-Term Target)
- **Low Accuracy / Low Effort:** Basic Retro-Reflection (Current Baseline)
- **The Sweet Spot:** Parallax/Viewpoint-Shift (Pure Software Win)

1. Ranked by Detection Accuracy

Methods are ordered from highest confidence to lowest baseline performance:

- **SweepLED (~94% Accuracy):** Best-documented performance baseline (30-object test evaluation). Explicitly engineered to filter out false positives that fool basic retro-reflection setups.
- **Thermal / HeatDeCam (High in Isolation):** Conceptually strong because operational electronics must dissipate heat. However, it degrades heavily in unconditioned environments due to ambient heat sources like lamps, sunlight, or warm surfaces.
- **Time-of-Flight / LAPD (Moderate):** Structurally limited by the coupled-motion ambiguity problem, where the light source and camera sensor move together on the same physical device axis.
- **Electromagnetic Emission (Highly Variable):** Bypasses optical camouflage entirely by sensing RF and electromagnetic signatures. However, signal strength drops fast with distance and is easily swamped by everyday household electronics.
- **Basic Retro-Reflection (Lowest Baseline):** High false-positive rates when encountering ordinary glossy, metallic, or reflective surfaces. This serves as the basic baseline performance indicator.

2. Ranked by Ease of Implementation

Ordered from immediate deployment readiness to heavy external hardware dependencies (assuming an ESP32 + mobile browser setup):

Rank	Method	Hardware Impact	Engineering Requirement / Notes
1	Basic Retro-Reflection	None (Existing BOM)	Zero. Current passive active pipeline is already built.
2	Parallax / Viewpoint Shift	None (Existing BOM)	Pure Software. Algorithm to track glint movement across frames as user moves phone.
3	SweepLED-style Sweep	Low (~\$7 LED Array)	Moderate. Hardware modification plus a trained lightweight verification model.
4	Polarization	Low (Physical Filter)	Moderate. Rotating filter addition combined with frame comparison logic.
5	Time-of-Flight (LAPD)	Restricted	High. Gated by high-end user phones equipped with LiDAR. Non-universal.
6	Thermal Imaging	High (New Sensor)	High. Requires adding an entirely new thermal sensor category to the current BOM.
7	EM Emission Analysis	Prohibitive	Extreme. Requires professional RF spectrum analysis hardware; well beyond ESP32 limits.

Strategic Action Plan

- The Immediate Win (Parallax Shift):** We should prioritize implementing the viewpoint-shift verification test next. It requires **zero new hardware additions**, utilizes our active camera stream, and significantly cuts down static false positives by evaluating glint stability.
- The Scaling Target (SweepLED):** If a high-confidence, commercially robust feature tier is required, SweepLED offers the most viable accuracy-to-cost scaling path (~\$7 added parts cost for a major jump to ~94% confidence).